Supplementary Materials: A Four-Dimension Pre-Modeling Audit Protocol for Educational Prediction Benchmarks

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This document contains supplementary tables and figures for the main manuscript.

S1 Supplementary Table

Table S1: Model-complexity ablation: instability ratio (I) and beat rate across linear and nonlinear models on 100 repeated 80/20 splits.

Dataset	Model	MAE (mean ± SD)	R^2 (mean)	r (mean)	I	Beat rate
MM-TBA	Linear	0.795 ± 0.089	−0.067	0.178	2.21	0.80
	Ridge	0.793 ± 0.089	−0.061	0.183	2.09	0.81
	Random Forest	0.815 ± 0.081	−0.063	0.196	3.96	0.65
	Gradient Boosting	0.856 ± 0.095	−0.227	0.129	4.60	0.45
Higher Ed	Linear	1.695 ± 0.199	0.041	0.469	0.90	0.83
	Ridge	1.686 ± 0.195	0.054	0.470	0.84	0.83
	Random Forest	1.193 ± 0.130	0.521	0.744	0.18	1.00
	Gradient Boosting	1.273 ± 0.146	0.464	0.711	0.23	1.00
xAPI-Edu	Linear	0.363 ± 0.025	0.599	0.786	0.12	1.00
	Ridge	0.357 ± 0.023	0.626	0.799	0.11	1.00
	Random Forest	0.308 ± 0.023	0.682	0.830	0.09	1.00
	Gradient Boosting	0.327 ± 0.024	0.663	0.818	0.10	1.00
Entrance Exam	Linear	0.598 ± 0.033	0.438	0.670	0.12	1.00
	Ridge	0.594 ± 0.032	0.448	0.676	0.11	1.00
	Random Forest	0.612 ± 0.036	0.372	0.628	0.14	1.00
	Gradient Boosting	0.595 ± 0.033	0.442	0.671	0.12	1.00
UCI Student	Linear	2.011 ± 0.148	0.262	0.530	0.36	1.00
	Ridge	2.010 ± 0.148	0.263	0.530	0.36	1.00
	Random Forest	2.005 ± 0.151	0.285	0.548	0.37	1.00
	Gradient Boosting	2.046 ± 0.151	0.257	0.531	0.41	1.00
Dropout	Linear	0.206 ± 0.006	0.651	0.807	0.021	1.00
	Ridge	0.206 ± 0.006	0.651	0.807	0.021	1.00
	Random Forest	0.153 ± 0.006	0.684	0.828	0.019	1.00
	Gradient Boosting	0.160 ± 0.006	0.692	0.832	0.019	1.00
OULAD	Linear	0.296 ± 0.002	0.471	0.687	0.010	1.00
	Ridge	0.296 ± 0.002	0.471	0.687	0.010	1.00
	Random Forest	0.193 ± 0.003	0.577	0.761	0.009	1.00
	Gradient Boosting	0.203 ± 0.002	0.610	0.781	0.008	1.00

S2 Supplementary Figures

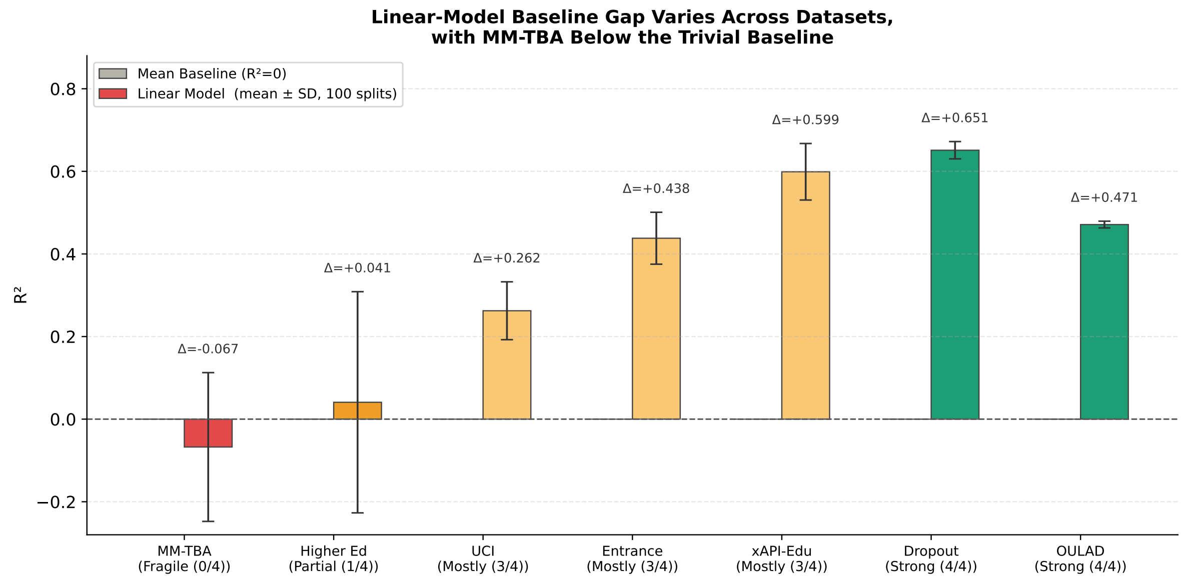

Fig S1: Baseline gap: linear-model mean R^2 (with cross-split SD error bars) over 100 random 80/20 splits for the seven datasets, ordered by readiness profile. The grey reference bar denotes the trivial mean baseline ($R^2 = 0$). The figure complements Table 2 by showing the dispersion of split-level R^2 around the reported mean; MM-TBA and Higher Ed have means near or below zero with relatively wide spread, consistent with their Fragile / Partial classifications.

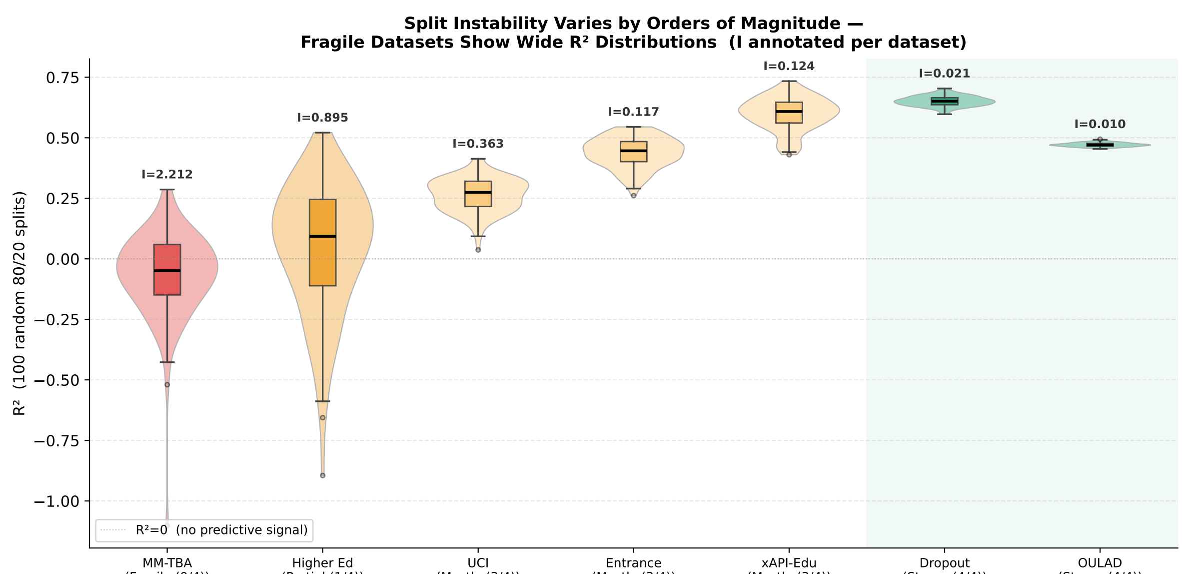

Fig S2: Split instability: violin and box plots of the linear-model R^2 across 100 random 80/20 splits for each dataset, ordered by readiness profile. The instability ratio I is annotated above each violin. Strong datasets (OULAD, Dropout) have tight distributions ($I < 0.03$); MM-TBA's distribution is broad and centred on negative R^2 ($I = 2.21$), reproducing Table 2 visually.

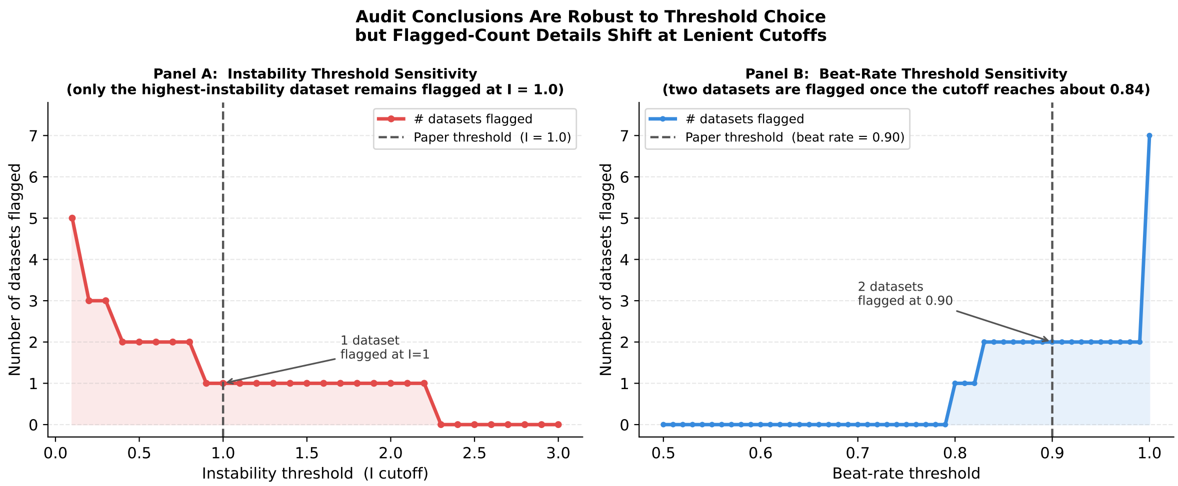

Fig S3: Threshold-sensitivity sweeps for the two operational cutoffs used in the audit. Panel (a) varies the instability cutoff over $I \in [0.1, 3.0]$ and reports the number of datasets that would be flagged as failing split instability; the paper's choice of $I = 1.0$ flags one dataset (MM-TBA, with linear $I = 2.21$), while more lenient cutoffs below about 0.9 additionally flag Higher Ed. Panel (b) varies the beat-rate cutoff over $[0.5, 1.0]$; the paper's choice of 0.90 flags two datasets, whereas more lenient cutoffs below about 0.84 reduce the flagged count. The main qualitative pattern remains unchanged: MM-TBA is always the weakest case, and Higher Ed is the only borderline dataset near the operational thresholds.